

VoiceMatchr: Final Project Submission Catalog

CS6460 Educational Technology, Summer 2026 - Development Track
Christopher Robinson Heydolph - cheydolph@gatech.edu

VoiceMatchr is a full-stack web application for self-directed voice-performance learning. A learner is able to select a synthesized target voice, submit a practice recording via audio file upload or live in-browser recording, and receives a speaker-embedding cosine distance to the target, a five-dimension acoustic comparison (F0 mean, F0 range, HNR, spectral tilt, loudness), template-generated coaching text targeting the largest gap, and a longitudinal progress chart across submissions. This catalog describes the contents of the submission archive and the externally hosted material.

Hosted Material

Live application	https://voicematchr.fiestaszn.com/
Source repository (public)	https://github.com/cheydolph/voicematchr

The live URL is served from a TrueNAS SCALE server host through a supervised Cloudflare Tunnel and requires no login to reach the onboarding flow. The tool runs on a server and a testing script is included (`scripts/smoke_test.py`, described below).

Archive Contents

Path	Description
Catalog.pdf	This document. Provides links to externally hosted material online.
README.md	README covering project overview, architecture table, quick start, Docker command reference, smoke test usage, test suite invocation, and evaluation data export instructions.
docker-compose.yml / nginx.conf / .env.example	Deployment configuration: the four-service compose stack (analysis service, frontend, nginx entry point, cloudflared tunnel connector) and the reverse-proxy configuration.
voicematchr-service/	FastAPI analysis backend: Resemblyzer speaker embeddings, openSMILE eGeMAPSv02 feature extraction, cosine-distance scoring, coaching text taxonomy and templates, SQLite repository layer, and the 37-test pytest suite.
frontend/	TanStack Start (React, TypeScript, Tailwind) learner application: onboarding with voice preview, browser recorder with waveform display, upload panel, radar chart, and longitudinal progress chart.
scripts/seed_prototypes.py	Idempotent voicebank seeding through the running API (standard library only).

Path	Description
<code>scripts/smoke_test.py</code>	Testing script for the hosted tool: checks service health, lists the registered voicebank, exercises Kokoro-backed preview synthesis, then onboards a marked test user and runs a full analyze round trip on a generated WAV. Run with no arguments to test the live URL: <code>python3 scripts/smoke_test.py</code>
<code>scripts/export_results.py</code>	Exports the evaluation dataset from the SQLite database as CSV: one row per analyzed recording with session, target voice, submission ordinal, cosine distance, per-dimension deltas, and coaching text.
<code>results.csv</code>	Session logs from the self-administered formative evaluation, produced by <code>scripts/export_results.py</code> .
<code>stack_implementation_guide.md</code>	Authoritative deployment and operations document (TrueNAS SCALE, Dockge, Cloudflare Tunnel, shared <code>kokoro-net</code> network), plus the Kokoro-TTS stack files (<code>kokoro-compose.yml</code> , <code>nginx-kokoro.conf</code> allowlist proxy).
<code>data/</code>	SQLite database and recordings volume layout (mounted at <code>/data</code> in the service container).

Running the Tool

The simplest evaluation path is the live URL above. To run locally: copy `.env.example` to `.env` and set `KOKORO_BASE_URL` to a reachable OpenAI-compatible `/v1/audio/speech` endpoint, then `docker compose build && docker compose up -d`, seed with `python3 scripts/seed_prototypes.py`, and open `http://localhost:8080/`. The full instructions, including the UID 1001 data-directory ownership prerequisite, are in `README.md` and the stack implementation guide.